

लोक सेवा आयोग
नेपाल राष्ट्र बैंक, प्राविधिक, सूचना प्रविधि, अधिकृत तृतीय, सहायक निर्देशक पदको खुला
प्रतियोगितात्मक लिखित परीक्षा
२०७९/०३/२९

KEY [A]

पत्र : द्वितीय
समय : ३ घण्टा

पूर्णाङ्क : १००

विषय : **Information Technology - II**

उत्तरपुस्तिकामा प्रश्नपत्रको **KEY** अनिवार्य रूपले उल्लेख गर्नुपर्नेछ । उल्लेख नगरेमा उत्तरपुस्तिका रद्द हुनेछ । परीक्षामा **calculator, mobile** प्रयोग गर्न पाइनेछैन ।

Section 'B' : Objective

Time : 30 mins

30x1=30 Marks

Multiple Choice

1. Which of the following options leads to the portability and security of Java?
A) Bytecode is executed by JVM
B) The applet makes the Java code secure and portable
C) Use of exception handling
D) Dynamic binding between objects
2. In how many places can variables be declared in C programming language?
A) One B) Two C) Three D) Four
3. Higher-order functions are not built into the
A) structural language B) object oriented programming
C) JAVA D) C++
4. Which of the following languages is translated with the interpreter?
A) C B) PASCAL C) COBOL D) BASIC
5. Which of the following data structure is non linear type?
A) String B) List C) Stack D) Graph
6. For an algorithm, the time complexity is given by $O(1)$ then complexity of it is
A) constant B) polynomial C) exponential D) none of the above
7. In which binary tree traversal will the output produce sorted key values in an ascending order?
A) In-order Traversal B) Pre-order Traversal
C) Post-order Traversal D) Both A and C
8. The postfix form of $(x+y)*z$ is
A) $xy*z+$ B) $xyz*+$ C) $xy+z*$ D) $xyz+*$
9. The complexity of Binary search algorithm is
A) $O(n)$ B) $O(\log_2 n)$ C) $O(n^2)$ D) $O(n \log_2 n)$
10. $p \rightarrow q$ is logically equivalent to
A) $\neg p \vee \neg q$ B) $p \vee \neg q$ C) $\neg p \vee q$ D) $\neg p \wedge q$

11. Which of the following can only be used in disproving the statements?
A) Direct proof
B) Contraposition proof
C) Counter example
D) Mathematical induction
12. If $A=\{a,b\}$, $B=\{2,3\}$ and $C=\{3,4\}$ then $A \times (B \cap C)$ is
A) $\{(a,2),(a,3),(a,4),(b,2),(b,3),(b,4)\}$
B) $\{(a,3),(b,3)\}$
C) $\{\varphi\}$
D) $\{a,b,3\}$
13. A top-down parser generates
A) rightmost derivation
B) rightmost derivation in reverse
C) leftmost derivation
D) leftmost derivation in reverse
14. The role of the lexical analyzer is to
A) produce the intermediate code
B) produce the output sequence of token
C) analyze the language grammar
D) generate the symbol table
15. Which of the following errors can be a compiler check?
A) Syntax error
B) Logical error
C) Both A and B
D) Compiler cannot check errors
16. Which one of the following models is not suitable for accommodating any change?
A) Build & Fix Model
B) Prototyping Model
C) RAD Model
D) Waterfall Model
17. Graphic representation of the control logic of processing functions or modules representing a system is known as
A) Structured Analysis
B) Structured Chart
C) Structured English
D) System Flow Chart
18. What is the major drawback of using RAD Model?
A) Highly specialized & skilled developers/designers are required
B) Increase reusability of components
C) Encourage customer/client feedback
D) Both A and B
19. Data cannot flow between two data stores because
A) it is not allowed in DFD
B) a data store is a passive repository of data
C) they will get merged
D) data can get corrupted
20. In which software process model is risk estimation considered?
A) Waterfall mode
B) Prototyping
C) Incremental development
D) Spiral model
21. What is the first stage in program development?
A) Specification and Design
B) System Analysis
C) Testing
D) Maintenance

22. Which of the following is not a maturity level in CMM?
A) Design B) Repeatable C) Managed D) Optimizing
23. Technical risks threaten the of the software.
A) System development B) Quality and timeliness
C) System integration D) All of the above
24. Post method of HTML form sends data through the
- A) html footer B) url C) http header D) html body
25. Which of the following is an essential process in which the intelligent methods are applied to extract data patterns?
A) Warehousing B) Data mining C) Text mining D) Data selection
26. Source code translation is a part of
- A) Data re-engineering B) Refactoring
C) Restructuring D) None of the above
27. Which is not an advantage of client-server computing?
A) It is easy to protect the data
B) The server need not be located physically close to the clients
C) It is easy to replace, upgrade or re-locate the nodes
D) Clients and server need to be built on similar platforms
28. In which of the following projections, the object size differs when looked from different distances?
A) Parallel projection B) Perspective projection
C) Cavalier projection D) Cabinet projection
29. Which one is an impact printer?
A) Electrostatic printer B) Line printer
C) Inkjet printer D) Laser printer
30. Raster graphics are composed of
- A) Pixels B) Paths C) Palette D) Vectors

- The End -

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Section 'A' : Subjective

Time : 2 hrs 30 mins

70 Marks

1. How do you compare procedural and object oriented programming? Write down the limitations of procedural programming. 3+2=5
2. Illustrate how binary search works. 5
3. Define Growth of Functions. How do you measure it? Explain with appropriate examples. 1+4=5
4. Design a lexical analyzer generator and explain it. 5
5. What is the purpose of the System Development Life Cycle (SDLC)? 5
6. List the different types of system testing. Explain why we need Software Configuration Management (SCM). 2+3=5
7. What is ERP? Explain its importance. 2+3=5
8. Define projection in computer graphics. Briefly explain different types of projections. 1+4=5
9. Write down the importance of inheritance in C++. Explain constructor and destructor call sequence in single and multiple inheritance in C++. How can we resolve ambiguity in multiple inheritance in C++?. 2+4+4=10
10. Compare object-oriented analysis and design with the structured analysis and design. Discuss different activities involved in each of the phases of the object-oriented development life cycle. 4+6=10
11. Discuss the importance of Software Quality Assurance (SQA) in brief. What are the types of software maintenance? Give some design principles for maintainability. 2+4+4=10

- The End -